

Y2 Assignment — Mobile Dashboard for RFID Wallet System

Dear Y2 students,

We are now moving the RFID Wallet Top-Up & Payment System from Web Dashboard → Mobile Dashboard.

Your objective is to replicate the same system functionality on a mobile device while keeping the same backend architecture.

Group Formation

You will form **groups of 6 students**.

Each group must include:

- 2 students from Y2A
- 2 students from Y2B
- 2 students from Y2C

This ensures cross-class collaboration.

Each group will receive access to a Lenovo tablet for testing.

Project Objective

Develop a Mobile Dashboard Application that performs the same tasks as the Web Dashboard you previously built.

The mobile dashboard will interact with the system through:

- MQTT communication
- Backend database
- RFID card system

The overall architecture remains the same.

RFID Reader (ESP8266 + MFRC522)



MQTT Broker



Backend Server



Mobile Dashboard

You may run your **MQTT broker locally on your PC** during development.

Required Mobile Features

Your mobile dashboard must support the following operations.

1) Agent Login — Wallet Top-Up

The mobile app must allow an **Agent** to:

- login to the system
- select or scan a card
- enter top-up amount
- confirm the transaction

The system must:

- update wallet balance in database
- update balance stored on the card
- record the transaction history

2) Salesperson Login — Payment

The mobile app must allow a **Salesperson** to:

- login
- select a product/service
- process payment using RFID card

The system must:

- deduct balance
- update the database
- generate a transaction record

3) Dashboard View

The mobile app must show a summary dashboard, including:

- number of cards
- total wallet balance
- number of transactions
- top-ups
- payments

MQTT Communication

Mobile app should communicate with the system using MQTT topics.

Example structure:

rfid/<team_id>/scan
rfid/<team_id>/topup
rfid/<team_id>/payment
rfid/<team_id>/status

Each group must use its *own team ID*.

Database Requirements

Your system must store:

- card UID
- wallet balance
- transaction type
- transaction amount
- timestamp
- terminal ID

Example record:

card_uid
transaction_type
amount
timestamp
terminal

Suggested Technology

You may choose any mobile development approach:

Option 1: Flutter

Option 2: React Native

Option 3: Android Studio (Java/Kotlin)

Option 4: Progressive Web App (PWA)

Deliverables

Each group must submit:

1) GitHub Repository

Repository must include:

README.md
mobile_app/
backend/
database/
mqtt_topics.md

2) Demonstration

During presentation you must demonstrate:

- ✓ Agent login
- ✓ Wallet top-up
- ✓ Sales transaction
- ✓ Balance update
- ✓ Dashboard statistics

Learning Goal

By completing this project you will learn:

- IoT mobile dashboards
- MQTT real-time communication
- RFID payment systems
- Mobile application development
- distributed system architecture